

A Solana Seeker app on the Subscriptions standard, for Clock In and the Crypto World's Fair

nuntius

Grant it once in Seed Vault.

The chain holds the line.

A capped, revocable, recurring payment you approve once, with a push receipt for every pull, including the ones the chain refused.

Problem

Solana has a recurring-payments standard. Users have no place to see or stop it.

Live since 2 June 2026

The Foundation's Subscriptions program. Audited by Cantina. The same address on devnet and mainnet.

u64::MAX at the token level

Every grant makes a program PDA the token account's delegate for the maximum amount. The per-period cap lives inside the program. nuntius caps the token approval at the lifetime total in the same transaction.

Every integration is merchant-side

Plans, billing, collection. We found no wallet or app that shows the user these permissions, alerts when money moves, or ends them in one tap.

Sources: the program repository and CHANGELOG (read at commit 299dfea); twins survey in RESEARCH.md. "Merchant-side only" is a search result, not a census.

Grant. Watch. Control.

Grant, in one sentence

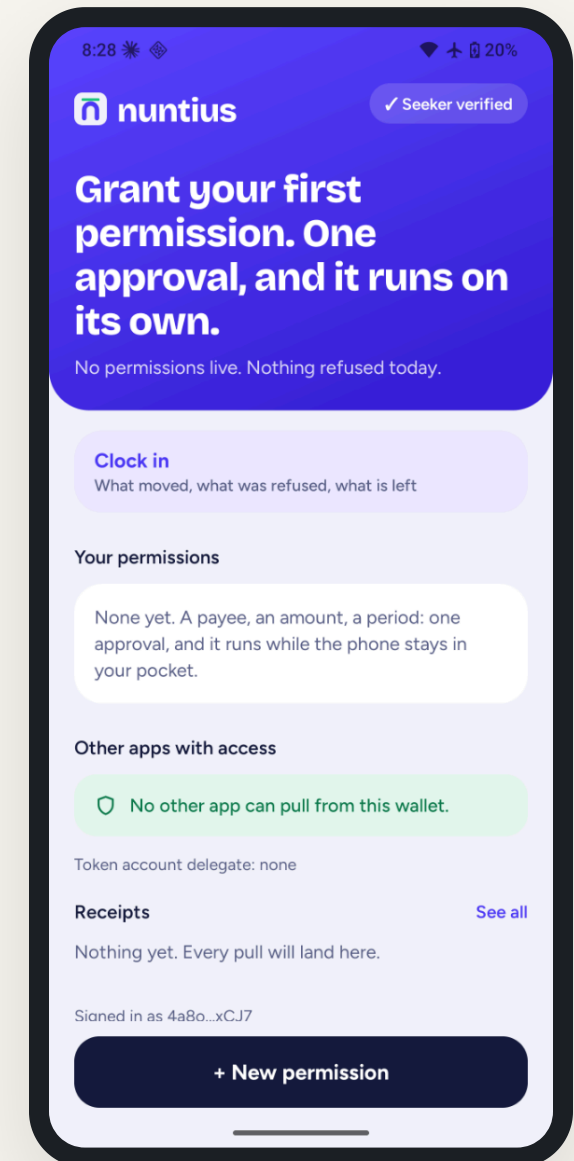
"Ana can receive up to 25 USDC every week, until 26 Dec." One Seed Vault approval. The first payment goes out at once, then one per period, with the phone in your pocket. Or type it your way, "Pay Ana 5 cents a day for a week", and Fill: a model reads it, plain code checks every term, you paste the payee. One tap backs a Seeker builder: 25 SKR a week.

Watch, with a receipt for every pull

Every pull opens the on-chain proof. When anything asks above the cap, the program refuses it (0x190) and the phone says so.

Control every permission

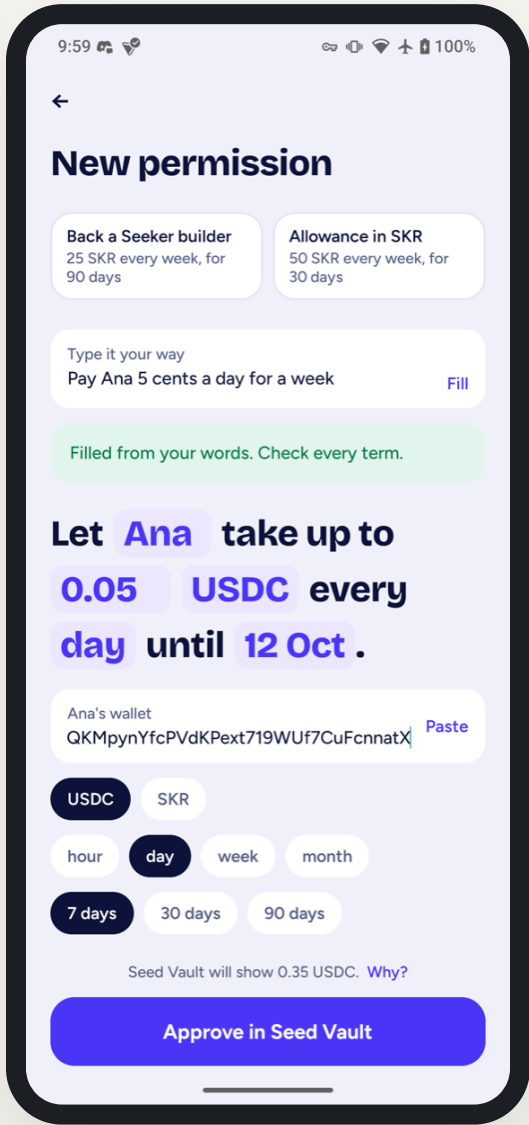
All Subscriptions delegations on the wallet, to nuntius or to any other app. Each ends with one approval.



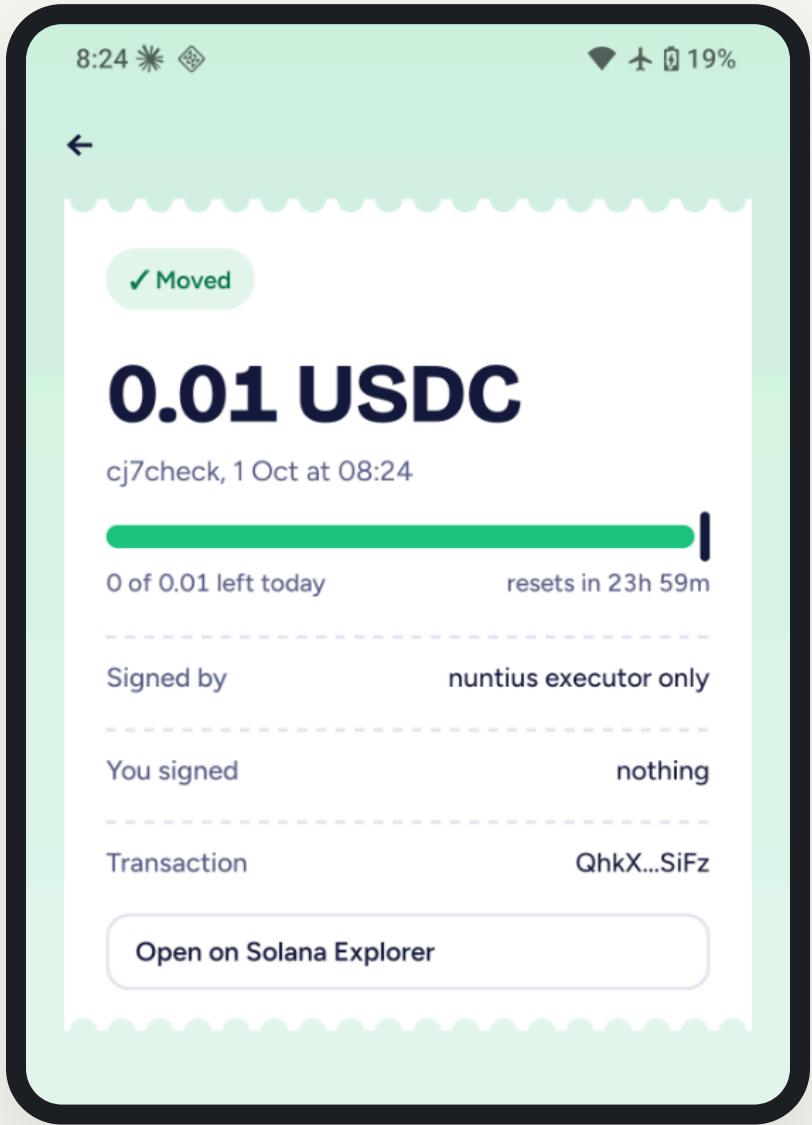
Home on the Seeker, mainnet, 1 Oct (round 3): the guard, the Seeker tier, and no delegate on the token account.

The phone

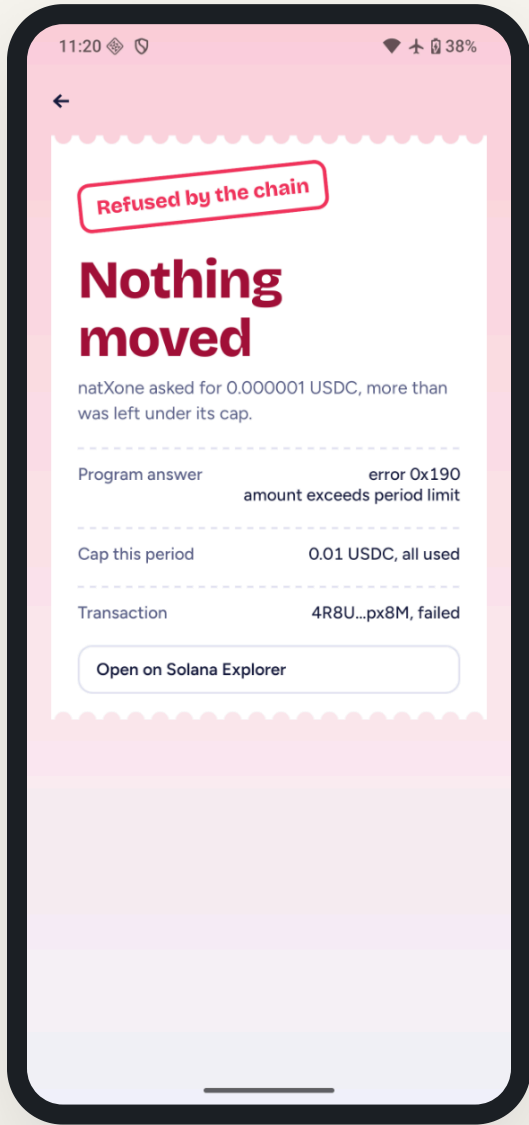
What the user sees



Typed in plain words, filled by Fill, every term checked by plain code. The payee is pasted by hand, and one line above Approve says what Seed Vault will show.



The "received" push, tapped with the app open, opens this receipt with its cap meter.



Refused by the chain, 0x190 on mainnet: the push, tapped with the app open, opens its own slip.

How one approval is enough

Authority and delegation in one transaction

initSubscriptionAuthority

Creates the (user, mint) authority PDA and makes it the token account's delegate. Idempotent when it already exists.

+

createRecurringDelegation

Amount per period, period length, expiry. It pins the authority's `init_id`, or `UNKNOWN_INIT_ID` when the authority is created in the same slot (program 0.5.0, #206).

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1 signer: the user

One Seed Vault sheet. Revoke is the same: `revokeDelegation` + `revokeSubscriptionAuthority` in one transaction when it is the last delegation on that mint.

Measured

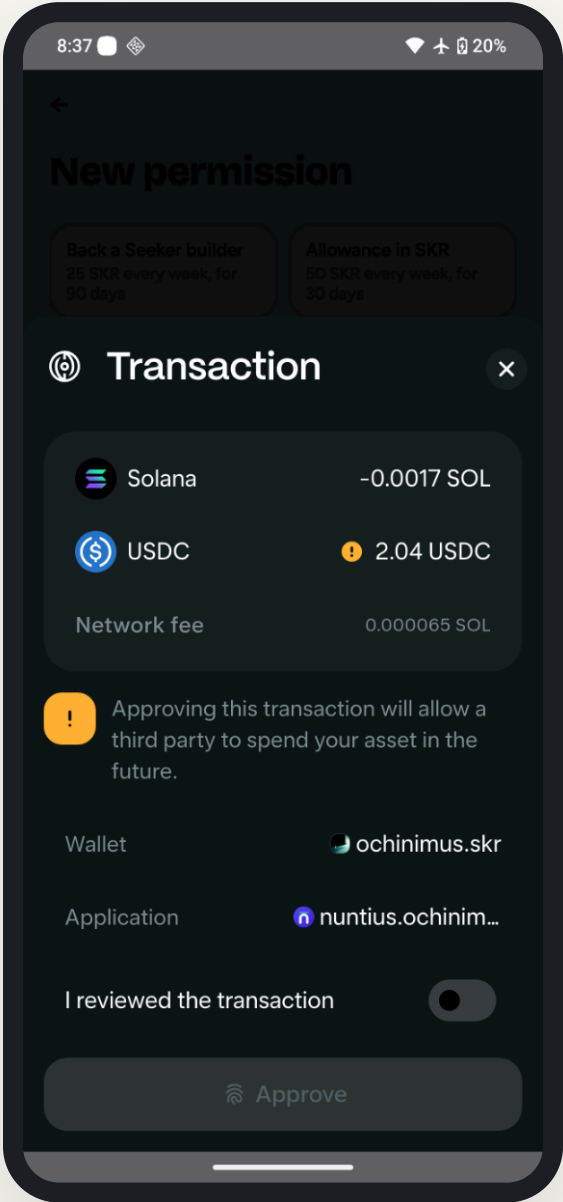
Four grants on mainnet from the Seeker, 30 Sep: each one Seed Vault sheet, each transaction signed by the owner alone. The localnet tests assert the same, and the exact terms.

The token approval is capped too

A third instruction, SPL `approveChecked`, lowers the allowance from `u64::MAX` to the lifetime total of the live permissions on that token. Next slide: what Seed Vault shows.

The allowance, measured on mainnet, 30 Sep – 1 Oct 2026 (rounds 2–3)

Seed Vault shows a number, not "Unlimited"



The third USDC permission: the app said 2.04 USDC, and so does Seed Vault.

Capped at the lifetime total

0.01 a day × 7 days: Seed Vault shows 0.07 USDC, not "Unlimited". On chain: delegate = the Subscription Authority, 0.07.

One approval, every live permission

With two USDC permissions live, the third grant's line said "Seed Vault will show 2.04 USDC in total", the sheet showed 2.04, and the chain read 2.03 after its first pull. "Why?" splits it: what this one can take, and the rest.

A revoke trims it

Revoking the hourly one set it to 0.30 (Seed Vault: "0.3 USDC"); the last revoke left de legat e: none. At the end, all 65 token accounts of both wallets had no delegate.

Mainnet, from the Seeker: round 4 on the release build 57eb4e1, 1 Oct 2026 (UTC)

Every tap lands, in every app state

Event	Result	Signature
Grant natXone 0.01 USDC a day, 08:16:29	one signer, the owner	2ibgJK5z...
First pull, 08:16:40	11 s after the grant; one push in the tray	4PSMS6nc...
Refused, tapped with the app open ×3, 08:20–08:23	0x190 each; every tap opened its own slip	4R8U6a5M... 2nbMHaUi... 5Nam7Lma...
Refused, tapped in the background, 08:23:46	0x190; the slip opened	4ZLE3GQF...
Refused, tapped from a cold start, 08:25:03	0x190; the slip opened, no crash	3uotigam...
Relaunch from the launcher	home, 3 of 3, also after a cold-start tap	—
Two revokes, 08:28:56 and 08:30:29	one sheet each; all 61 token accounts: no delegate	DZa2Vyzt... 2zZDAhNH...

Round 3 (same day): first pulls 4–7 s after their grants; the explainer matched Seed Vault at 2.04 USDC; no clipped text after a reboot; receipts per permission; four revokes, one sheet each. Round 2 (30 Sep): the SKR pull at 5,961 compute units and 7,000 lamports. Full signatures with Explorer links: README.md, "What is proven".

Proof: the real program, built from its release commit

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tests pass against **solana-foundation/subscriptions @ 364a419**, the commit its CHANGELOG names as the mainnet release, in a local validator (Agave 3.1.10).

`scripts/localnet.sh` builds it from source; the binary's hash depends on the build machine

- One-signature grant; second mandate reuses the authority with the real `init_id`
- Delegatee-only pulls; over-cap and one-unit-over refused with `{"Custom":400}`
- Executor: pulls, period roll and a second pull, a real 0x190, notices a revoke
- Guard: receipts for a foreign delegatee's pulls and refusals; flags new grants
- HTTP API end to end, driven exactly as the app drives it
- Revoke of the last delegation leaves `delegate: none`
- Two mints side by side, the SKR path: a ceiling per mint, one authority each

Also: 89 server unit tests (Type it your way with a mocked model: ceilings, unknown tokens, past dates, an injection, an address in the answer, a timeout), 46 app-logic tests (including the transaction check and the fill), 12 tests on the web build (cold-start tap, fonts, layout, BACK, launch hidden, the builder starter); typecheck, lint and prettier clean. Whether the local binary is byte-identical to mainnet has not been measured yet.

An executor that cannot double-pull

Idempotent per period

One ledger row per (delegation, period start), holding the signature *before* sending. A replacement is built only once the old blockhash can no longer land.

0x190 is final

Recorded as a refusal, pushed once, never retried. If the period's cap is already used, nothing is sent: no fees spent on a known no.

Flaky RPC, gone users

Exponential backoff with jitter. A closed delegation means revoked; past expiry means over. Each ends with one receipt.

16 unit tests on a simulated program (lost transactions, restarts, a send that throws after landing, max attempts, every send simulated first). Logs are JSON with keys and tokens redacted.

It comes to you, every day

A receipt on every pull

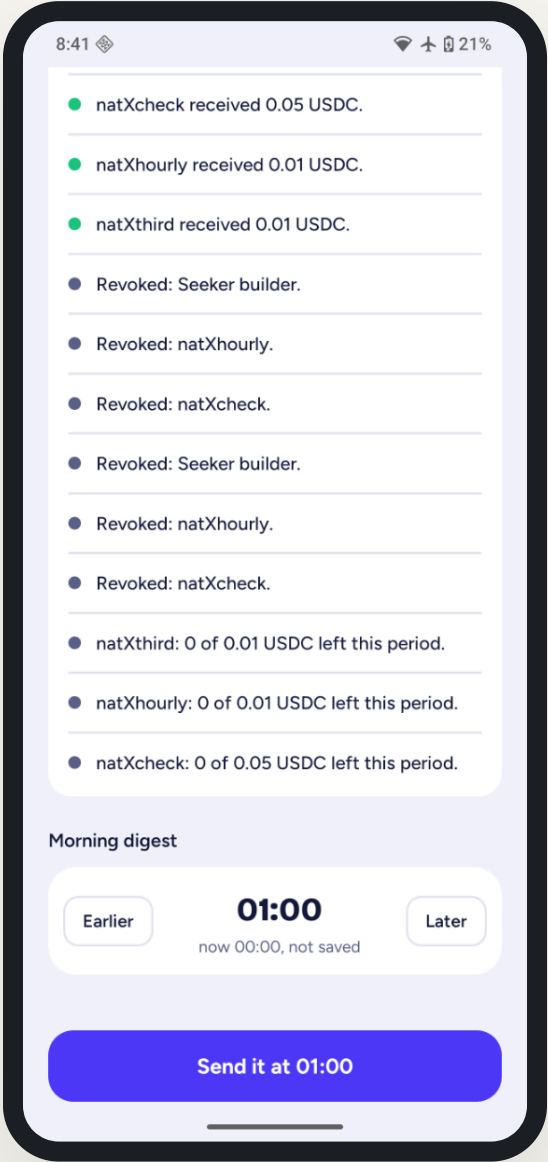
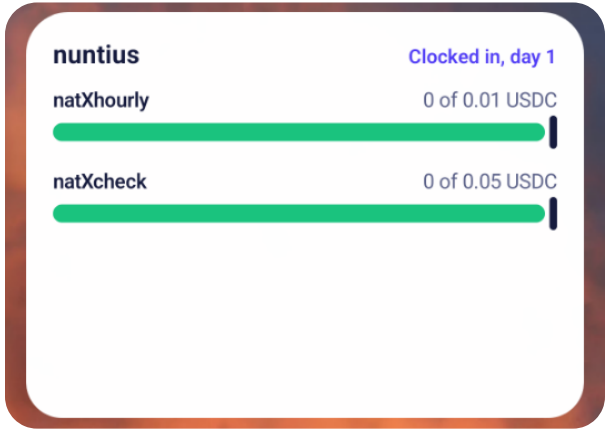
Money moves while the app is closed. The push is the reason to look.

Morning clock-in

At the hour you choose: what moved since the last digest, what was refused, what is left. Opening it is the clock-in, and it keeps a streak.

Home-screen widget

Cap left and time to reset for each permission.



Clock in on the Seeker, 1 Oct: an unsaved hour turns the bottom bar into "Send it at 01:00".
The widget (left) is from 30 Sep.

What the cap bounds, and what it does not

Bounded by the chain

Amount per period (0x190, the program) and the lifetime total (SPL Token allowance); the delegatee is the only signer; whole-period resets; hard expiry; revoke closes it. Our server cannot raise any of it.

Not bounded by the cap

The token approval is capped at the **lifetime total**, so an **upgrade** of the program reaches at most what you approved in total. Another app's grant can lift the cap. The **destination** is bound by nuntius, not by the chain.

Audit scope

The one-transaction path (#206) is inside Cantina's audited baseline d6b3a5dc. The release commit is 8 commits past the audit-fix commit.

This codebase

The app checks every server-built transaction before Seed Vault opens. The model behind Fill only fills the form: it never signs, never picks an address, and plain code checks each term against the server's limits. No secrets in the repo; SIWS nonces are single-use; sign-out ends the session on the server. npm audit: 0 on the server; the app's 5 are node-forge in Expo's build tools, not in the shipped bundle. Full table in SECURITY.md.

Field

Nobody else ships this on Seeker

	Standard audited program	Per-period cap	One Seed Vault approval	Receipt per pull	Other apps' permissions	Mainnet
nuntius	yes	yes	yes (mainnet, 30 Sep)	yes, incl. refused	yes	yes (22 Sep – 1 Oct)
Veto (Clock In)	own Anchor program	total and daily limits	Seed Vault owner key	on-chain refusal records	no	devnet
Kin (Clock In)	plain SPL approve	per round	MWA	reminders	no	devnet
Loonie	yes	yes	web wallet	no	no	devnet
Explorer card	yes	read-only	—	no	read-only	yes

From each project's public README, read on 27 Sep 2026 (RESEARCH.md). nuntius on mainnet: the 22 Sep life cycle (two approvals) and four device rounds, 30 Sep – 1 Oct (one approval per grant, receipts, refusals, widget, digest).

Where it stands, honestly

Done

v1.0.2 is done. Installed over v1.0.1 on 5 Oct, it filled "Pay Ana 5 cents a day for a week" into 0.05 USDC a day, matched Seed Vault at 0.35 USDC and pulled in 14 s. Back a Seeker builder took 25 SKR a week, matched Seed Vault at 325 SKR and pulled in 3 s. Both were revoked, and the first attempt's wallet error is now checked on chain. 130 tests against the real program.

Before submission

Clock In submission by 9 Oct. Crypto World's Fair submission by 12 Oct.

After

dApp Store listing. Talk to the merchants on the standard. A disclosed fee on mandated flow, with the guard free for everyone. No users or revenue yet.

Grant once. The chain holds the line.